

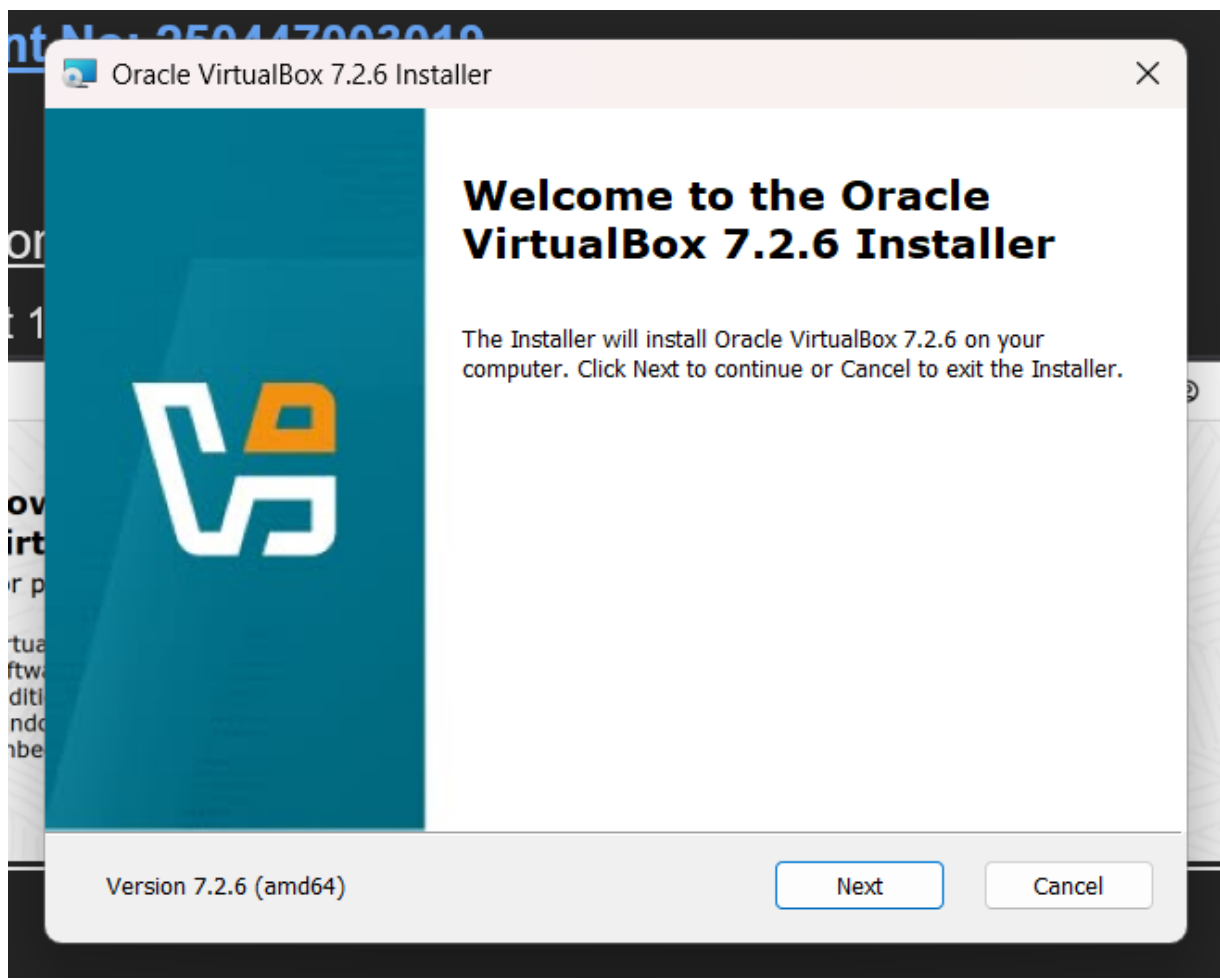
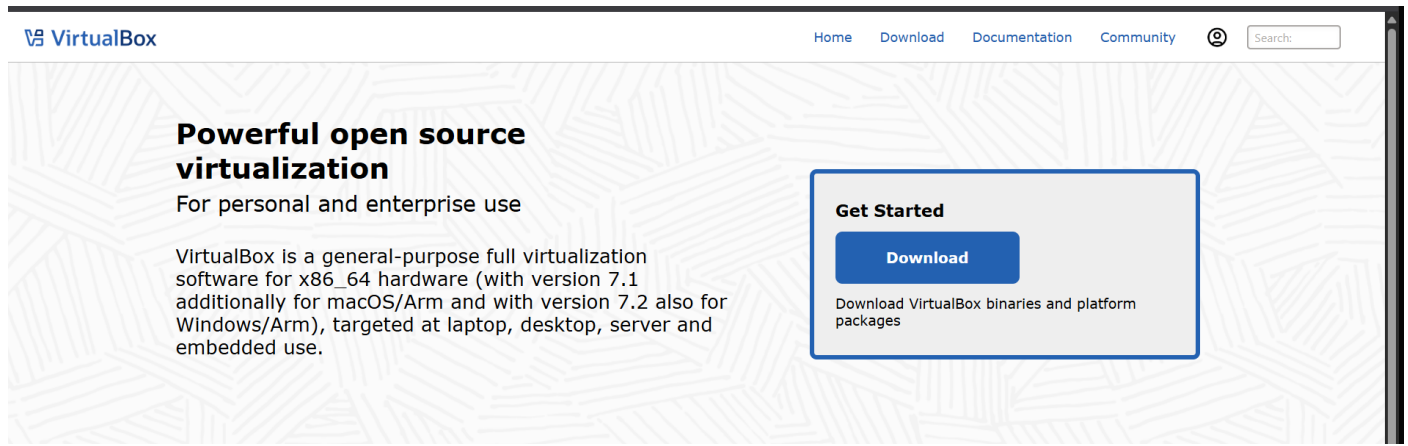
Name: Varia Neel Jitesh

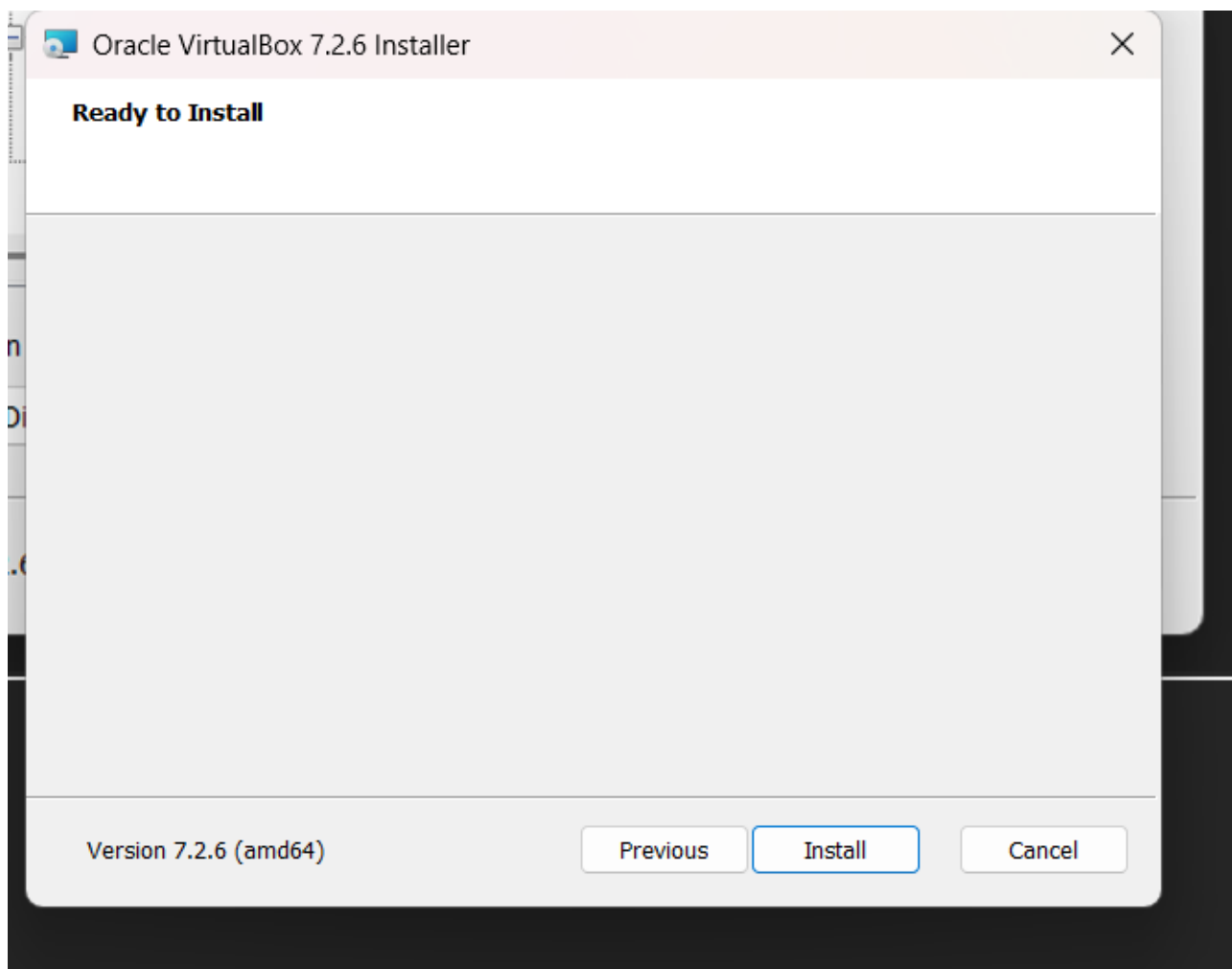
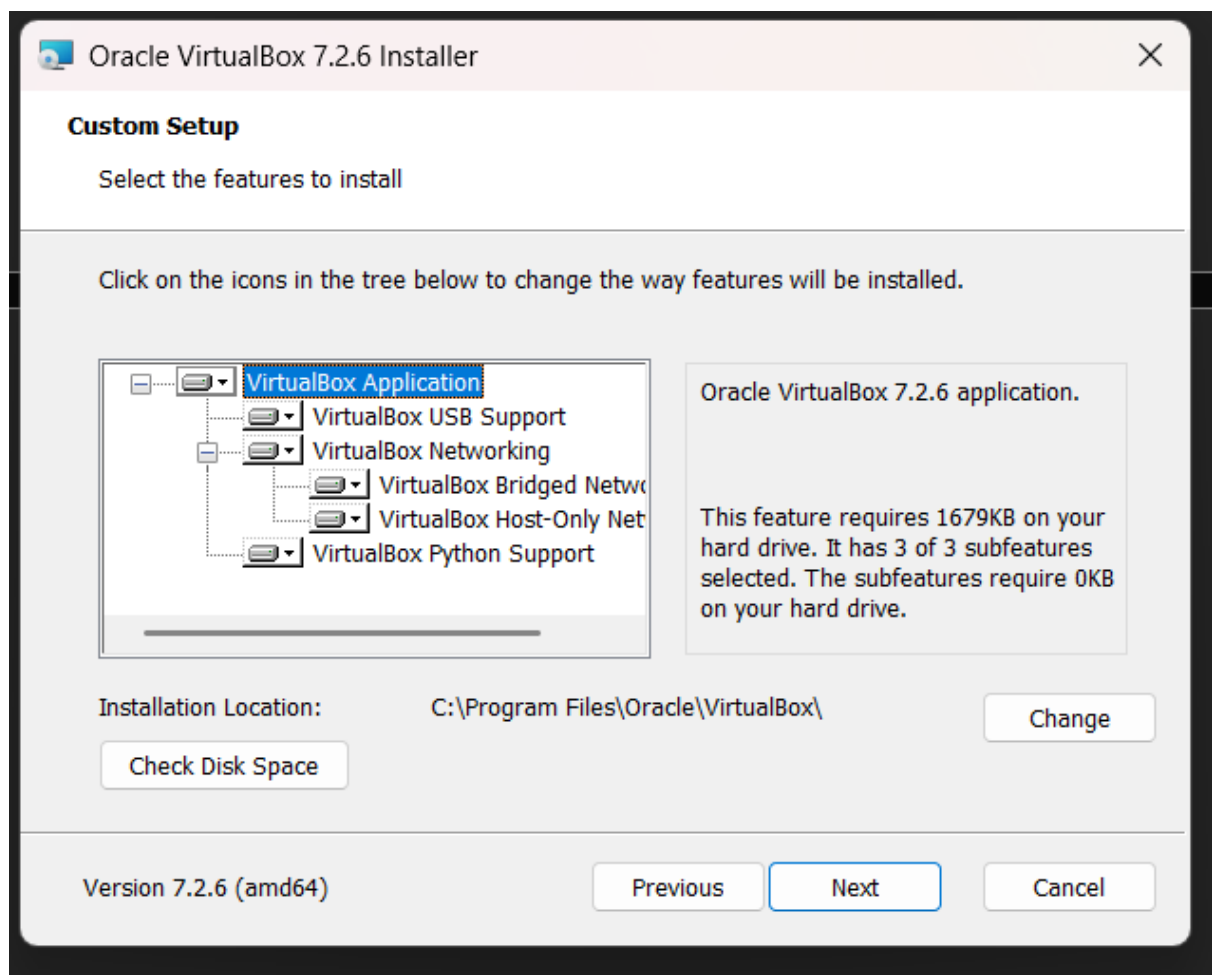
Enrollment No: 250447003019

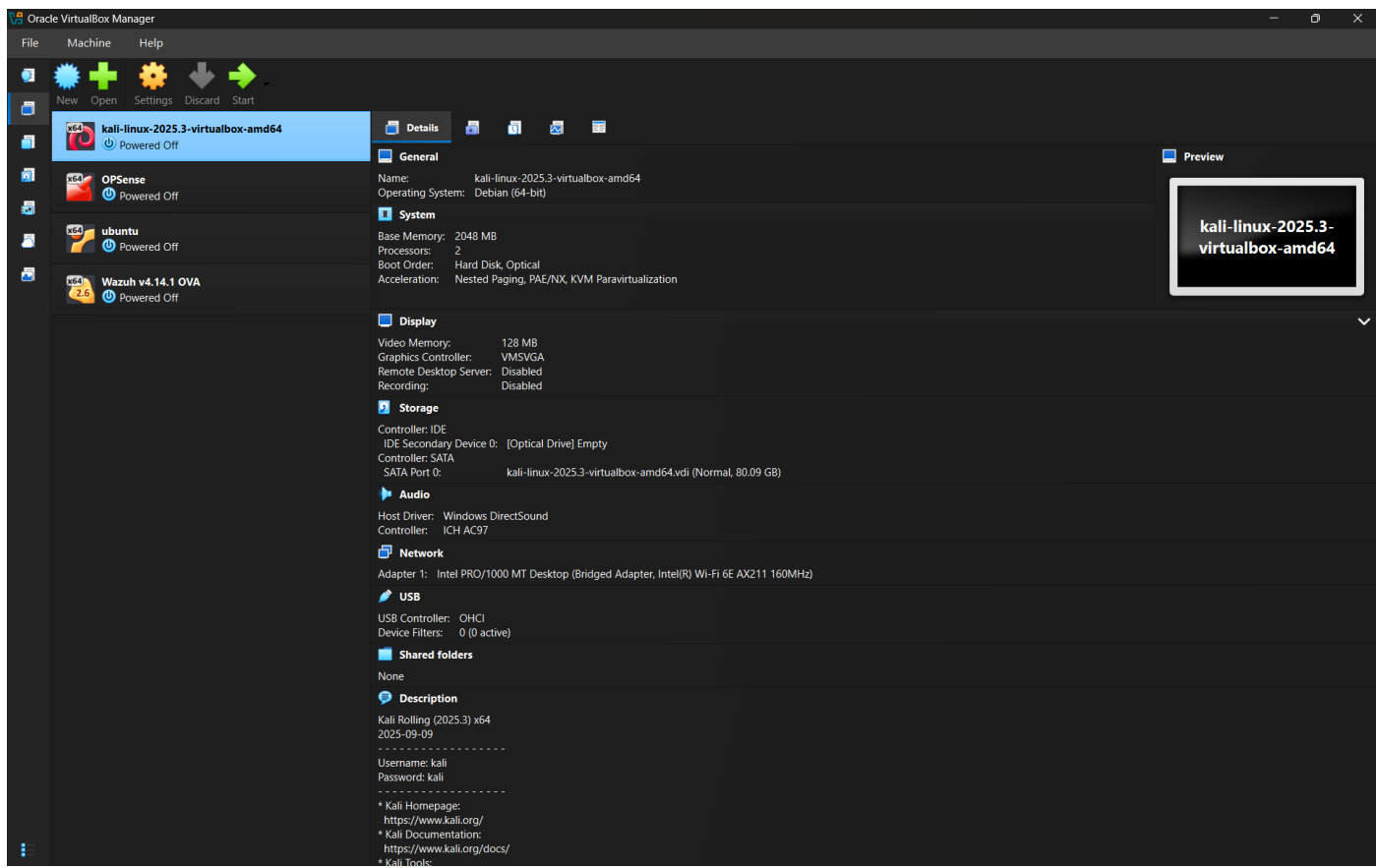
Network Security And Forensics

Lab Session – 1

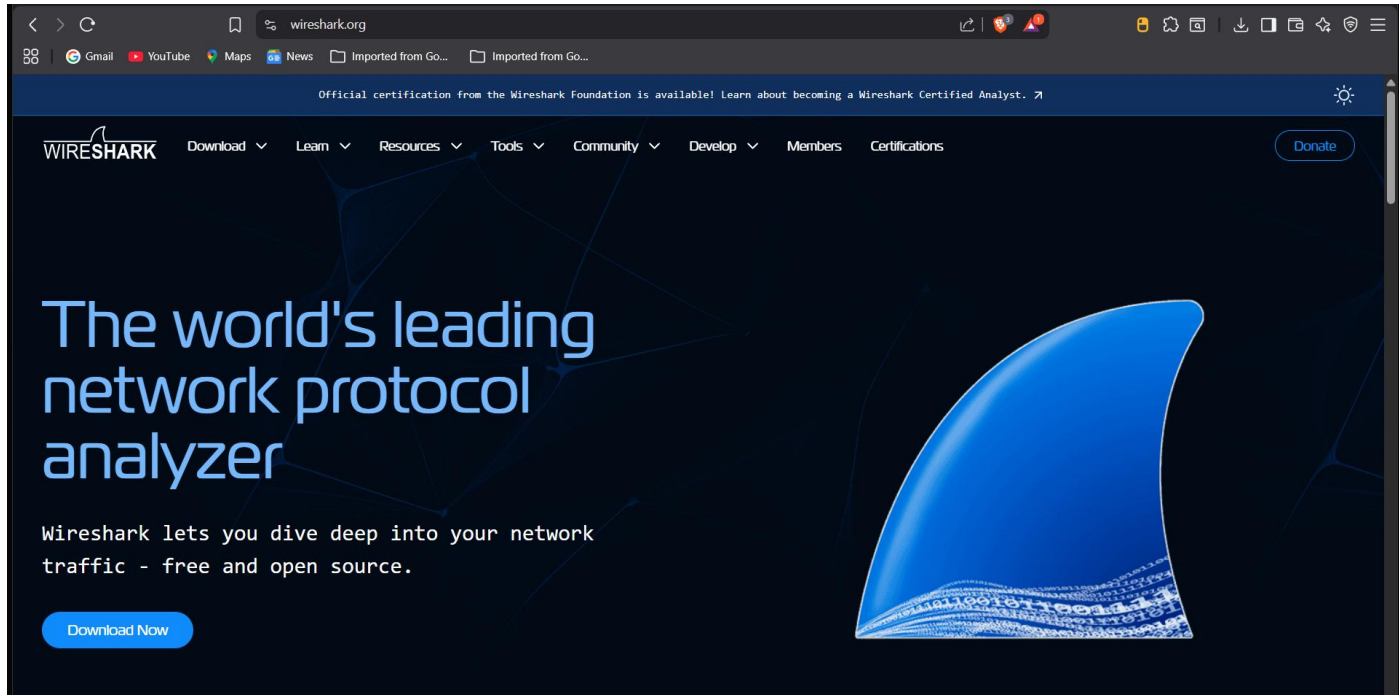
Assignment 1: Install VMware/ VirtualBox Workstation on Windows

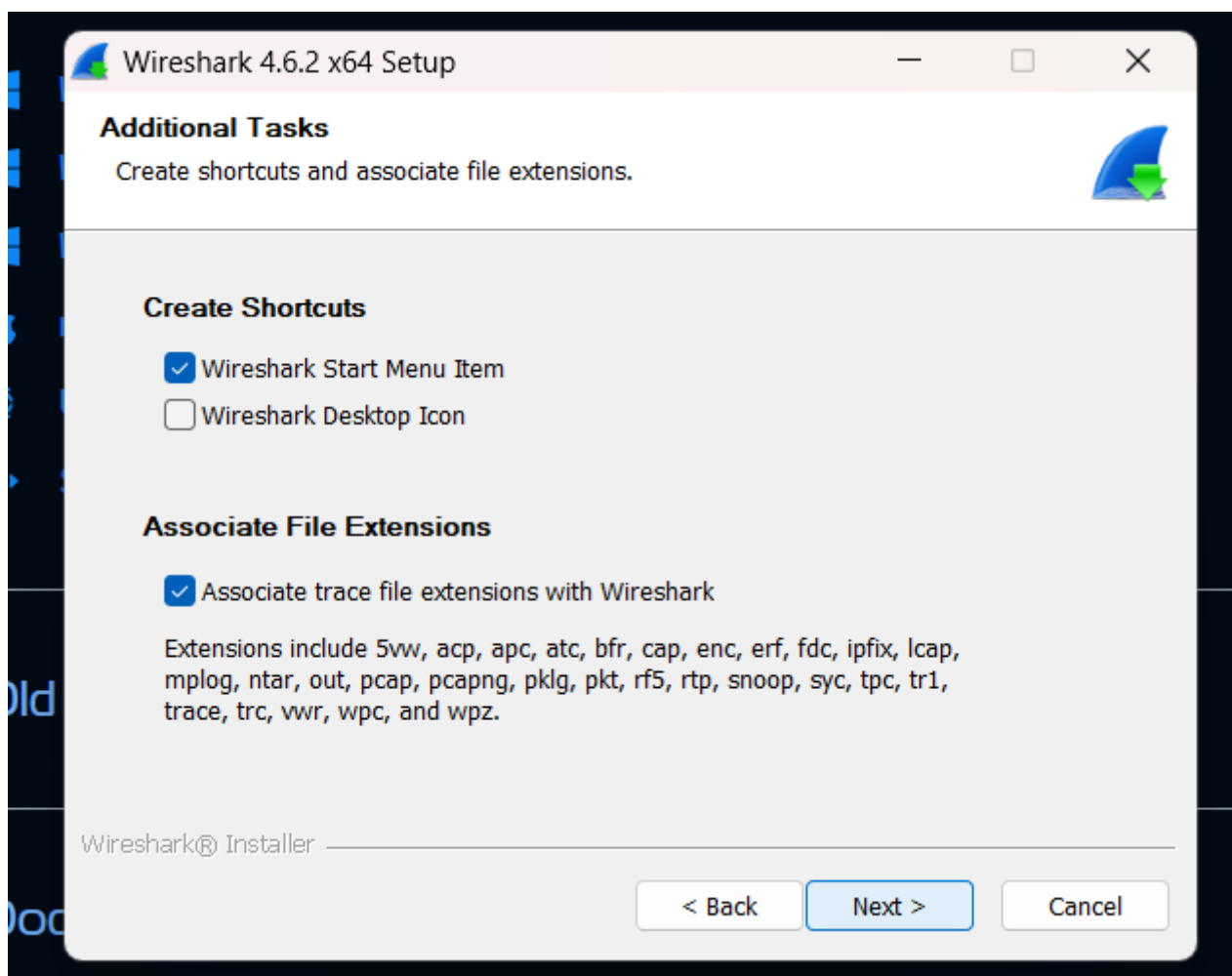
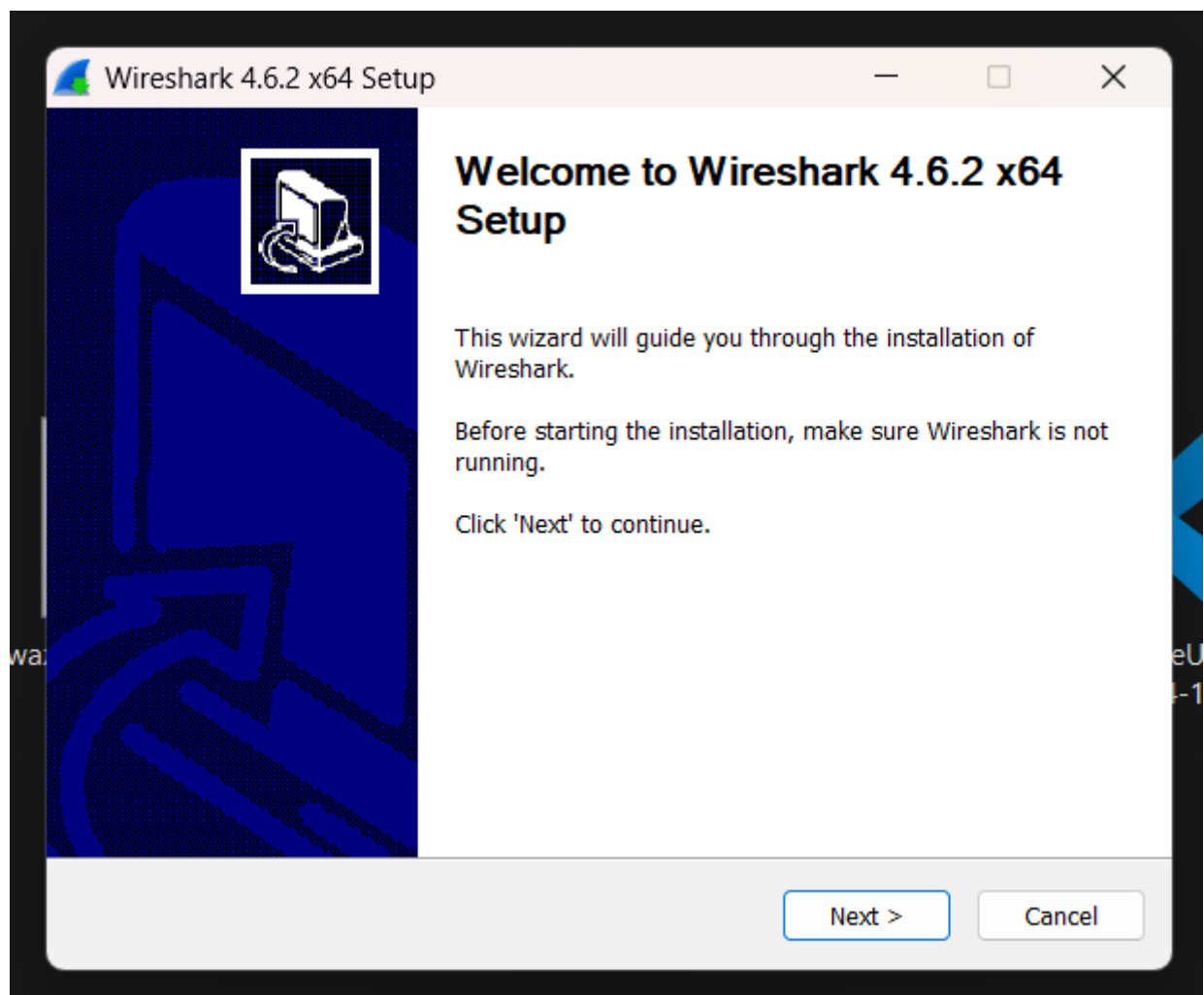






Assignment 2: Installation of Wireshark on Windows





ark 4.6.2 x64 Setup

al T Wireshark 4.6.2 x64 Setup



Completing Wireshark 4.6.2 x64 Setup

Wireshark 4.6.2 x64 has been installed on your computer.

Click Finish to close Setup.

☐ Open the release notes

< Back

Finish

Cancel

Installer

Assignment 3: Installation of Cisco Packet Tracer

computerNetworkingNotes.com/ccna-study-guide/download-packet-tracer-for-windows-and-linux.html

ComputerNetworkingNotes

Home CCNA Study Guide Linux Tutorials Networking Tutorials Career Resources

Download Packet Tracer 9.0.0 and all Previous Versions

The article provides direct download links to all versions (9.0.0, 8.2.2, 8.2.1, 8.2.0, 8.1.1, 8.0.0, 7.3.1, 7.3.0, 7.2.2, 7.2.1, 7.2.0, 7.1.1, 7.1.0, 7.0, 6.3, 5.0 & 4.0), platforms (Linux, Windows, and MAC), and editions of Packet Tracer with various downloading options (zip, exe, torrent, individual file or all files).

Packet Tracer is a network simulator software. It simulates networking devices used to build CCNA practice labs. An average CCNA lab costs more than \$300. Packet Tracer cuts this cost by simulating all essential CCNA lab devices. A simulation-based learning environment helps students and instructors. Students can learn how to configure routers and switches from the command line. Packet Tracer provides visual drag-and-drop problems using virtual networking devices. Instructors can use Packet Tracer to create their own self-evaluated assignments for students or demonstrate networking concepts.

Packet Tracer 9.0.0 (Latest Version)

[Cisco Packet Tracer 9.0.0 download link for Windows \(11, 10, 8.1\) 64 bits edition](#)

[Cisco Packet Tracer 9.0.0 download link for Linux 64 bits edition](#)

[Cisco Packet Tracer 9.0.0 download link for MAC os](#)

[Cisco Packet Tracer 9.0.0 torrent download link for Windows 64 bits, MAC, and Linux 64 bits](#)

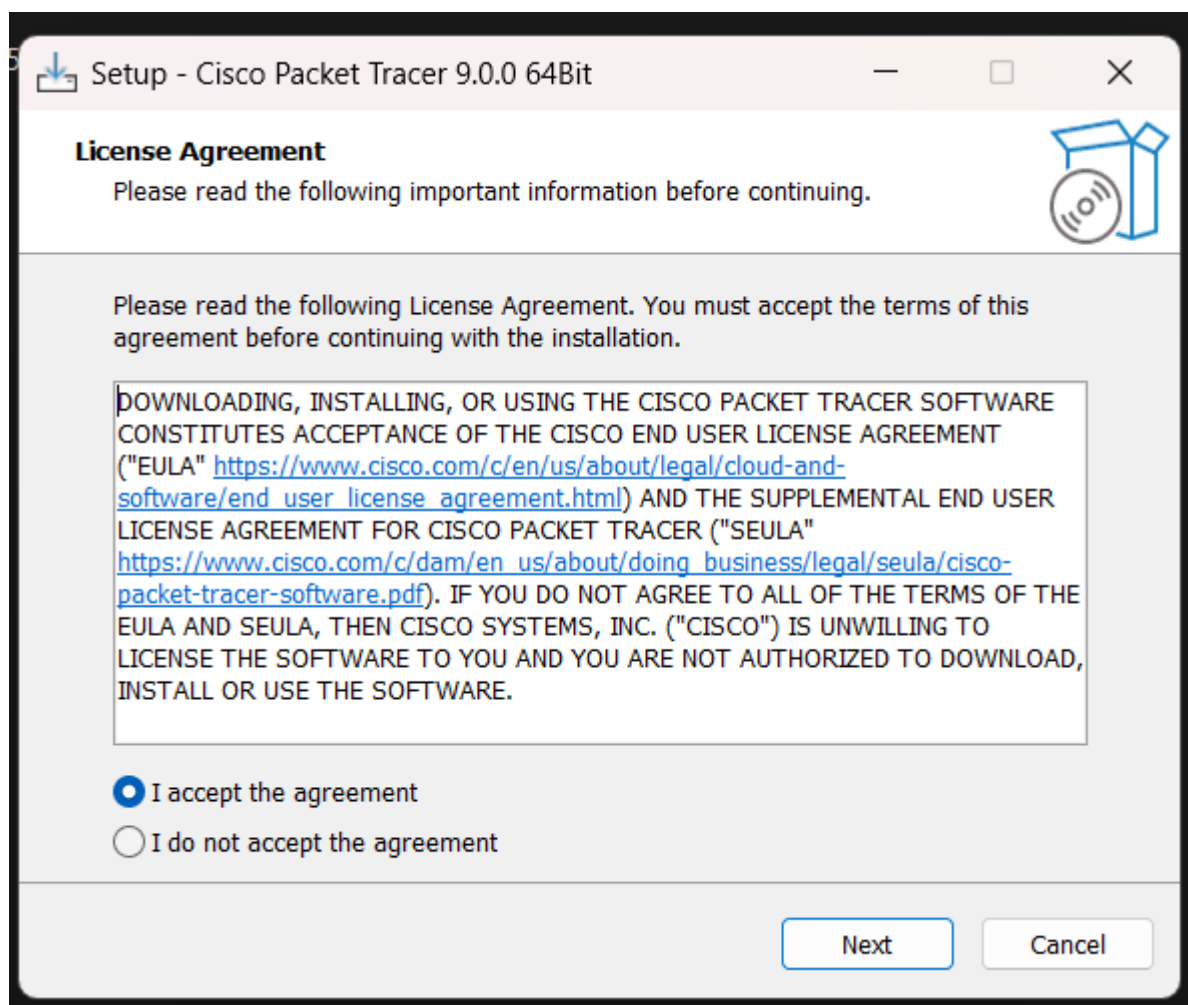
Switching concepts explained with examples

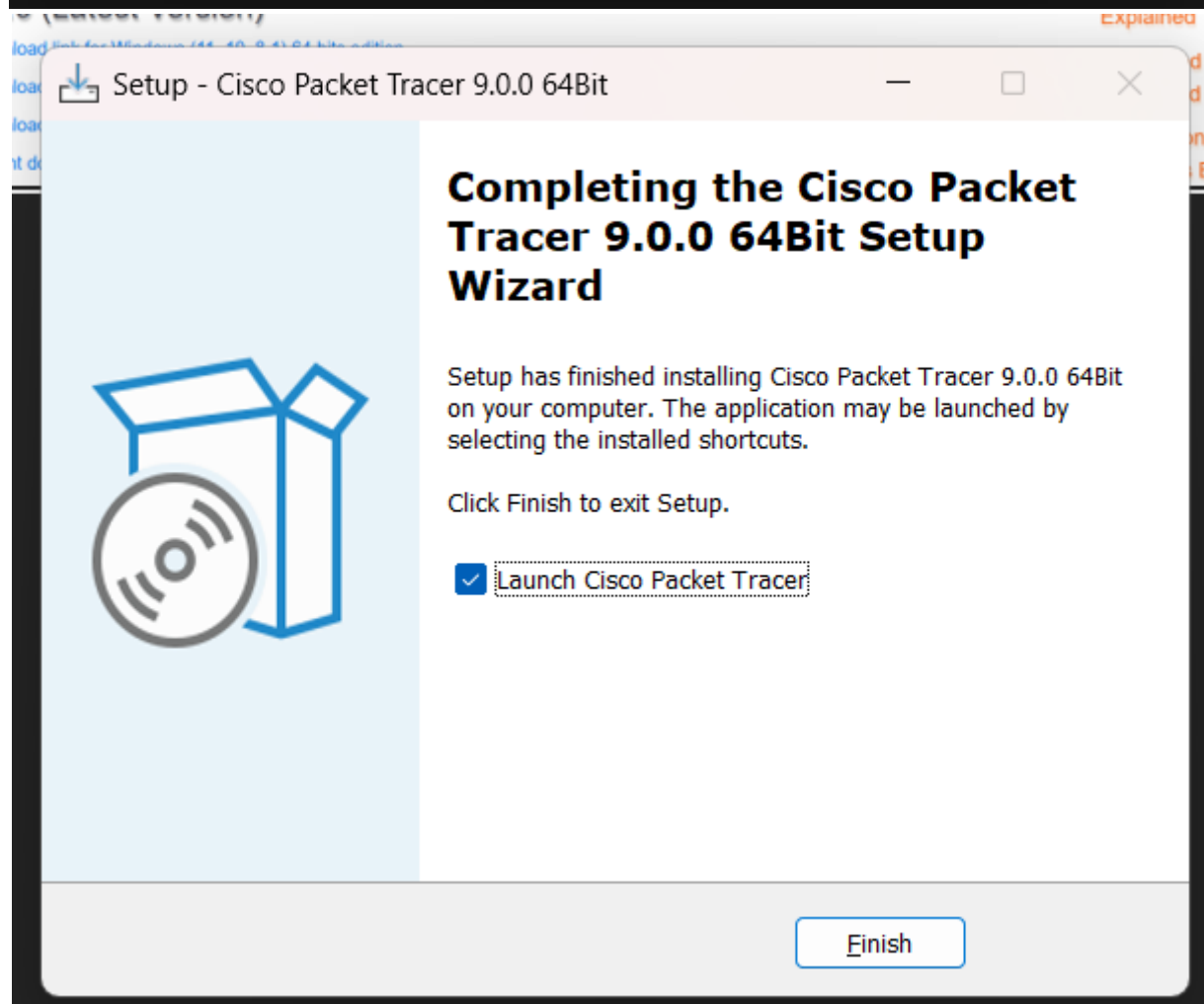
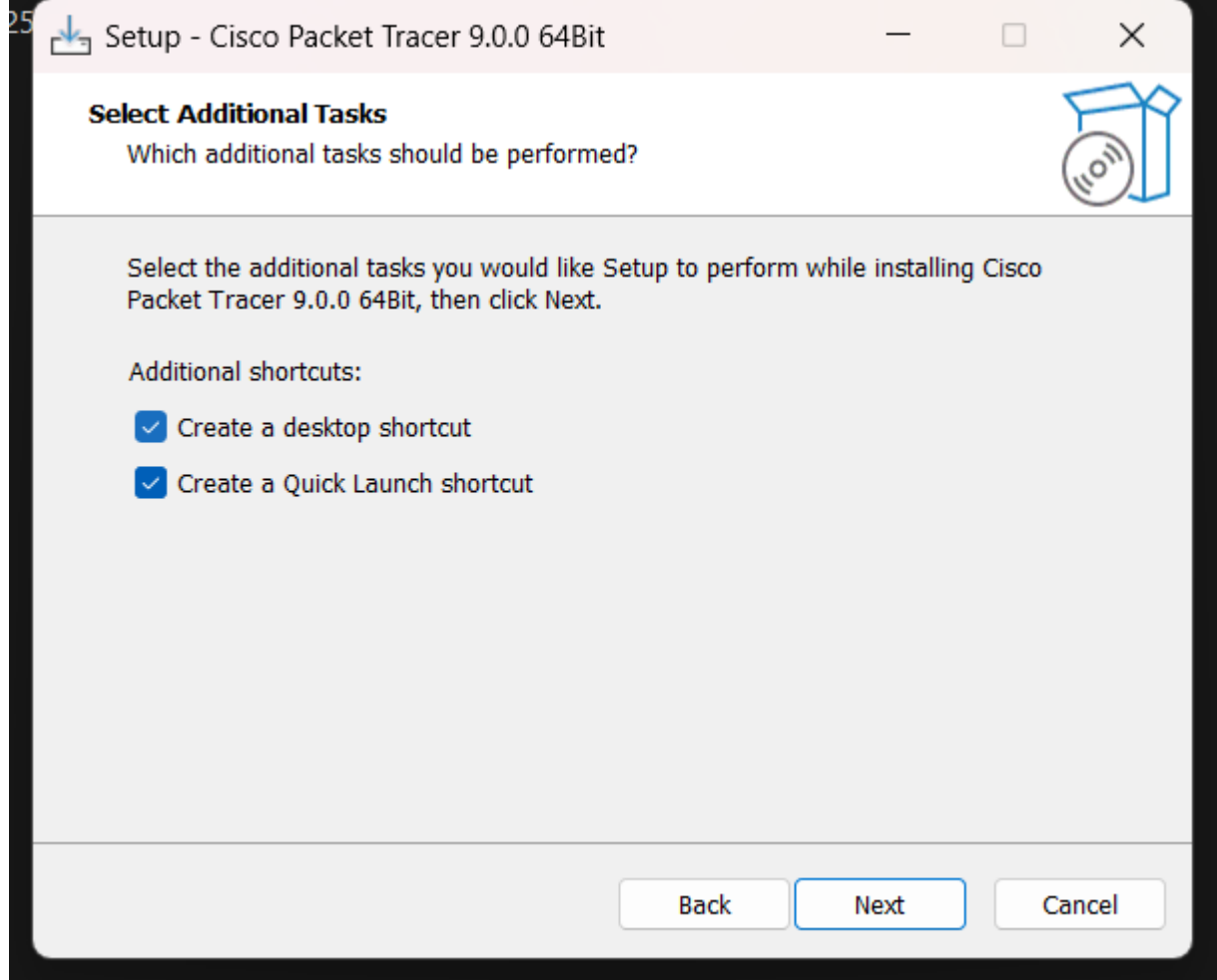
Basic Concepts of an Ethernet Switch

OSPF Neighborhood Building Requirements and States Explained

Virtualized Intel VT-x-EPT is not Supported

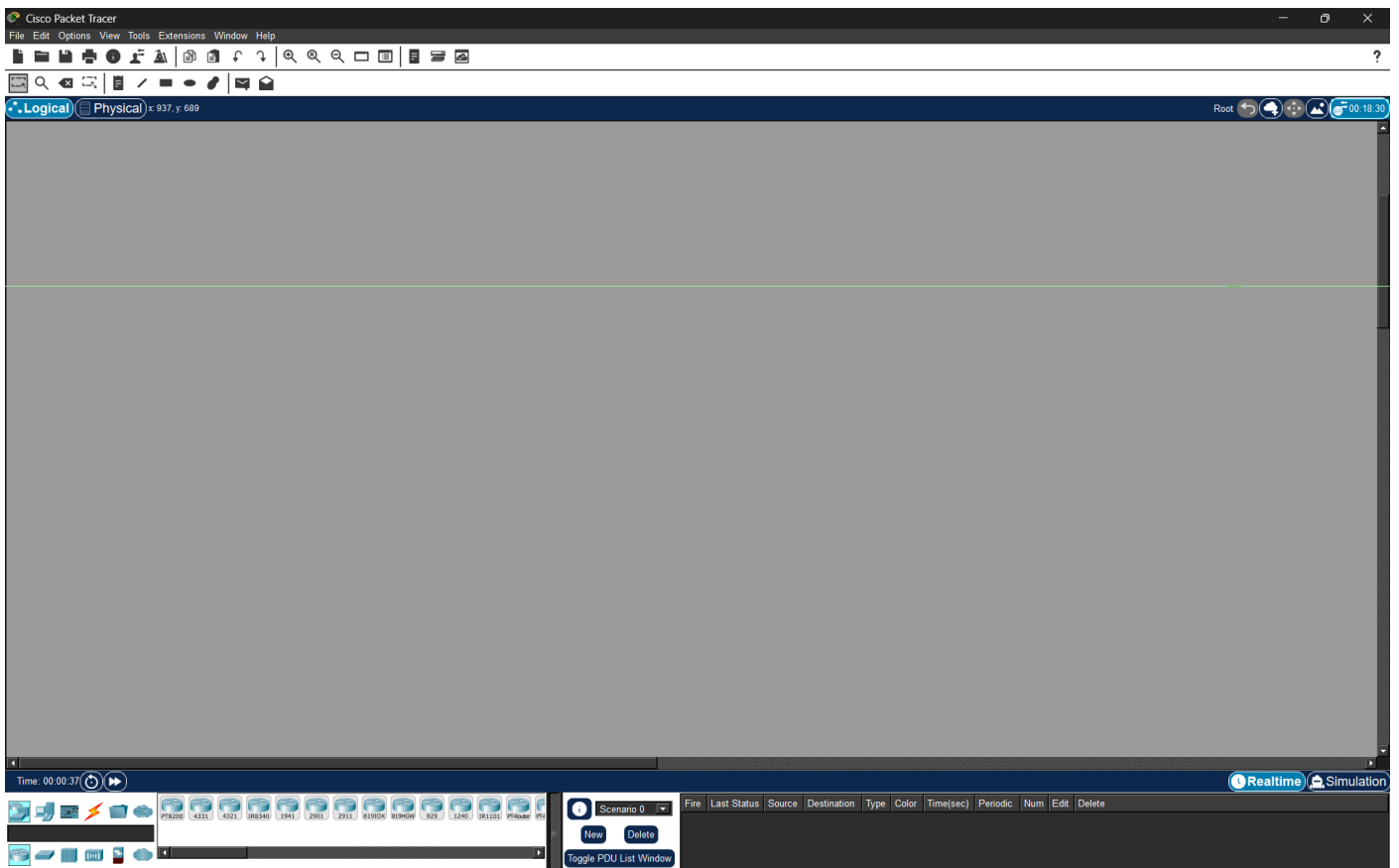
OSPF Configuration and Concepts Explained





CISCO PACKET TRACER

Program your own IoT smart solution.



Assignment 4: Installing NetworkMiner

netresec.com/?page=NetworkMiner#networkminer-download

OSINT lookups of file hashes, IP addresses, domain names and URLs

Port Independent Protocol Identification (PIPI) (**)

User Defined Port-to-Protocol Mappings (decode as)

Export to CSV / Excel / XML / CASE / JSON-LD

Configurable file output directory

Configurable time zone (UTC, local or custom)

Offline IP-to-Country lookup

Offline IP ASN lookup

DNS Whitelisting (***)

Advanced OS fingerprinting

Web browser tracing (video tutorial)

CIDR filter

Advertisement and tracker detection

Host coloring support

Command line scripting support

Price

Free

\$ 1300 USD

Download NetworkMiner (free edition)

Buy NetworkMiner Professional

* Fingerprinting of Operating Systems (OS) is performed by using databases from [Satori](#) and [p0f](#)

** Identified protocols include: DNS, FTP, HTTP, HTTP2, IRC, Meterpreter, NetBIOS NameService, NetBios

*** Whitelisting of domains, IP addresses, and file hashes

NetworkMiner_3-1

Downloads > network > NetworkMiner_3-1 > NetworkMiner_3-1

Search NetworkMiner_3-1

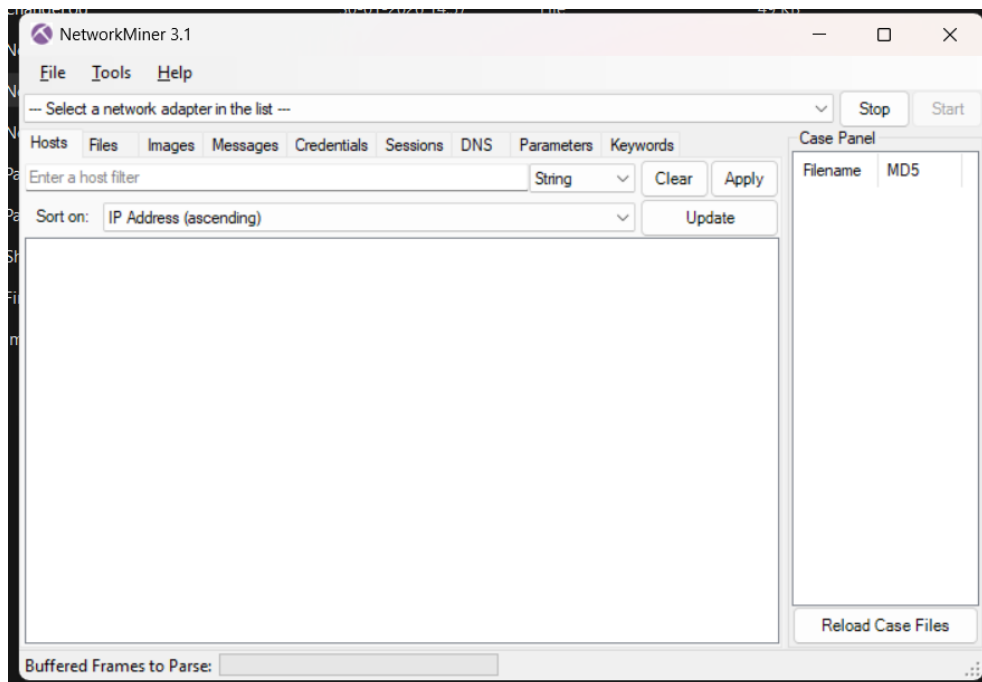
New

Sort

View

Details

Name	Date modified	Type	Size
Today			
ChangeLog	30-01-2026 14:57	File	49 KB
NetworkMiner.desktop	30-01-2026 14:57	DESKTOP File	1 KB
NetworkMiner	30-01-2026 14:57	Application	766 KB
NetworkWrapper.dll	30-01-2026 14:57	Application extens...	39 KB
PacketHandlerFramework.dll	30-01-2026 14:57	Application extens...	401 KB
PacketParser.dll	30-01-2026 14:57	Application extens...	549 KB
SharedUtils.dll	30-01-2026 14:57	Application extens...	92 KB
Fingerprints	30-01-2026 14:57	File folder	
Images	30-01-2026 14:57	File folder	



Assignment 5: Installation of ELK

Download Elasticsearch

Copy to try locally in two minutes: start-local

For local testing, use Docker to install and run Elasticsearch and Kibana on your local machine using start-local.

```
curl -fsSL https://elastic.co/start-local | sh
```

For production installations, follow the steps below.

1 Download and unzip Elasticsearch

Choose platform:

Windows

Windows

sha

asc

Package managers:

Summary

Version: 9.2.4

[View past releases](#)

Release date: January 13, 2026

[Detailed release notes](#)

Supported OS/JVM/Browser

[Support Matrix](#)

Notes:

Running on Kubernetes? Try [Elastic Cloud on Kubernetes](#).

This default distribution is governed by the Elastic License, and includes the [full set of features](#).

Elasticsearch

Discover

Dashboards

Agents

Machine Learning

Data management

My Elasticsearch proj... Getting started

AI Agent NG

Getting started

Start building with Elasticsearch.

Dive into an API tutorial or connect your deployment to start building.

Add data

Skip and go to Home

Elasticsearch endpoint: ⓘ

https://my-elasticsearch-project-ae55cb.es.us-central1.gcp.elastic.cloud:443

View connection details

Explore the API

Choose a tutorial and use Console to quickly start interacting with the Elasticsearch API.

Search basics

Learn how to create an index, add documents, and basic search techniques.

Intro to semantic search

Learn semantic search techniques to understand intent and deliver more accurate, relevant results.

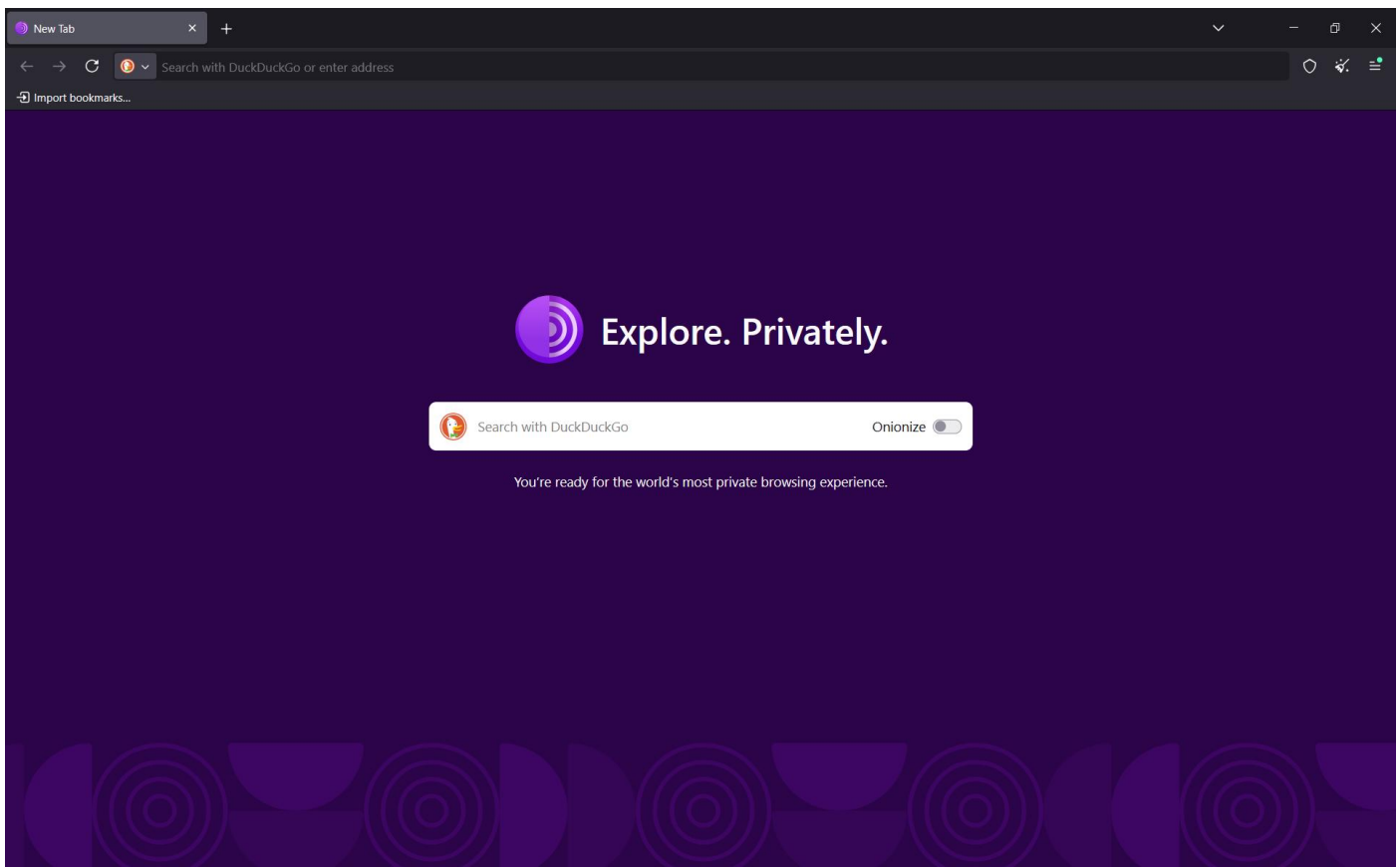
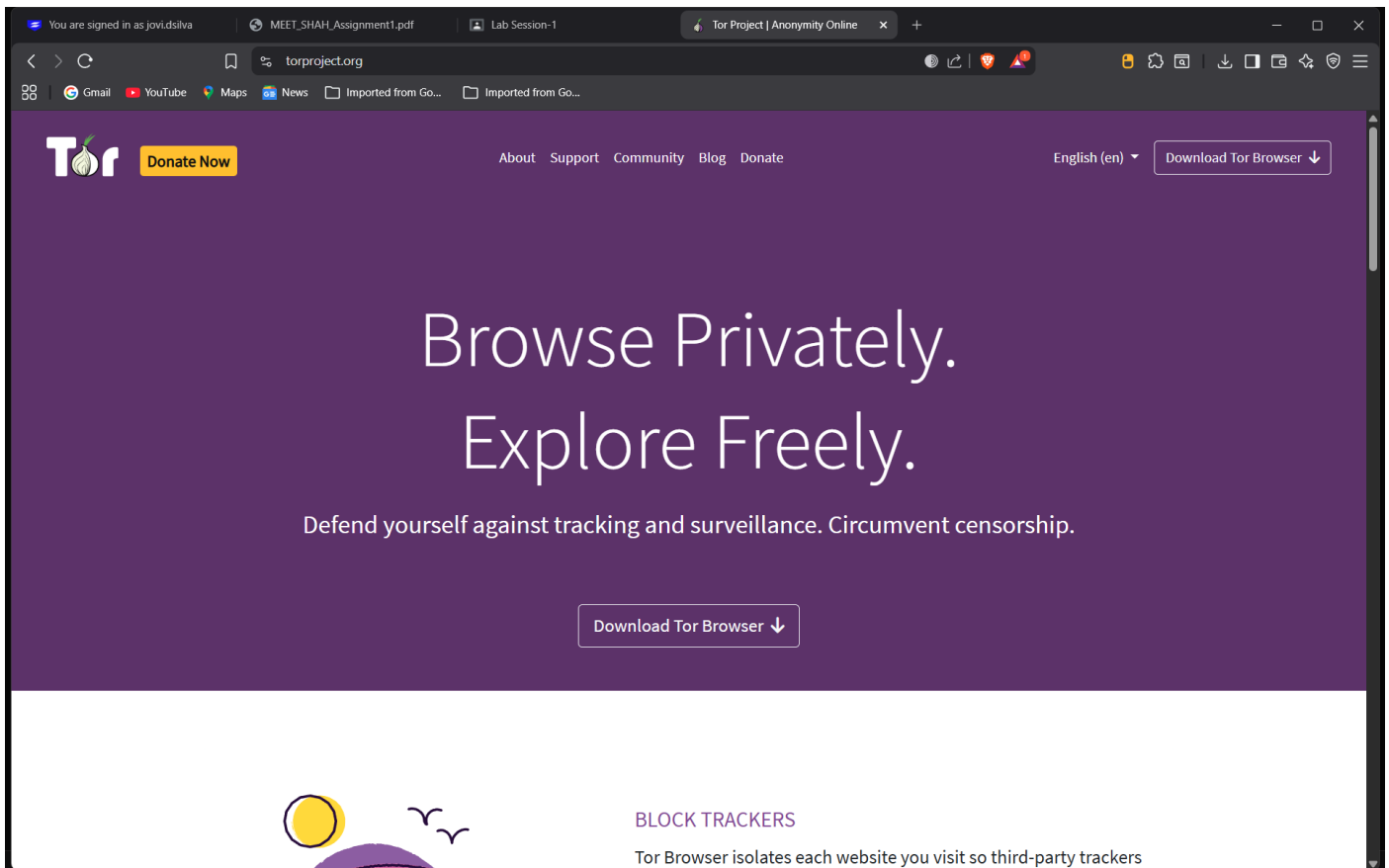
ES|QL fundamentals

Learn how to use Elastic's piped query language to simplify data investigations.

Assignment 6: Install and Perform various settings in following

Tor Browser Brave Browser:

Explore safe, anonymous browsing





The browser that puts you first

Block ads. Save data. And get way faster webpages. Just by switching your browser.

Get Brave ↓

Private search

Search the web privately...



Better results, AI answers, and zero profiling with [Brave Search](#).

Built-in AI assistant

Get answers, generate content, & more. Right in the browser.

Leo AI



Powerful VPN

Protect every app, on your entire device.

Firewall + VPN



User Account Control

Do you want to allow this app to make changes to your device?



BraveSoftware Update Setup

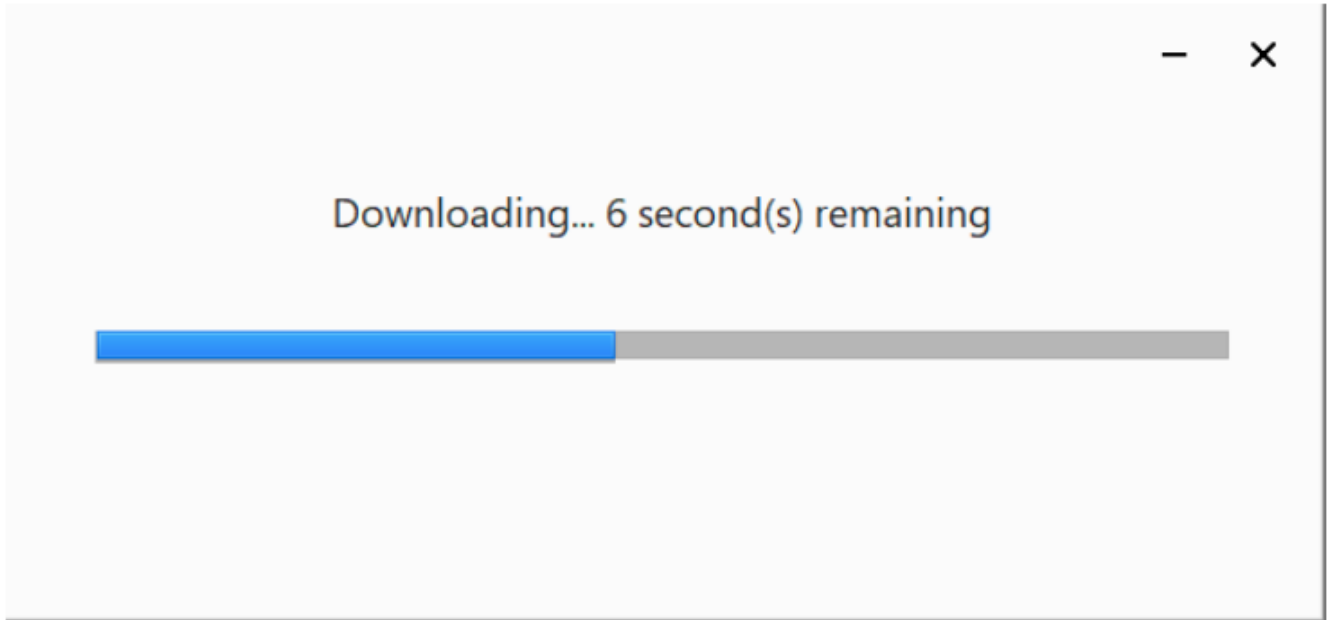
Verified publisher: Brave Software, Inc.

File origin: Hard drive on this computer

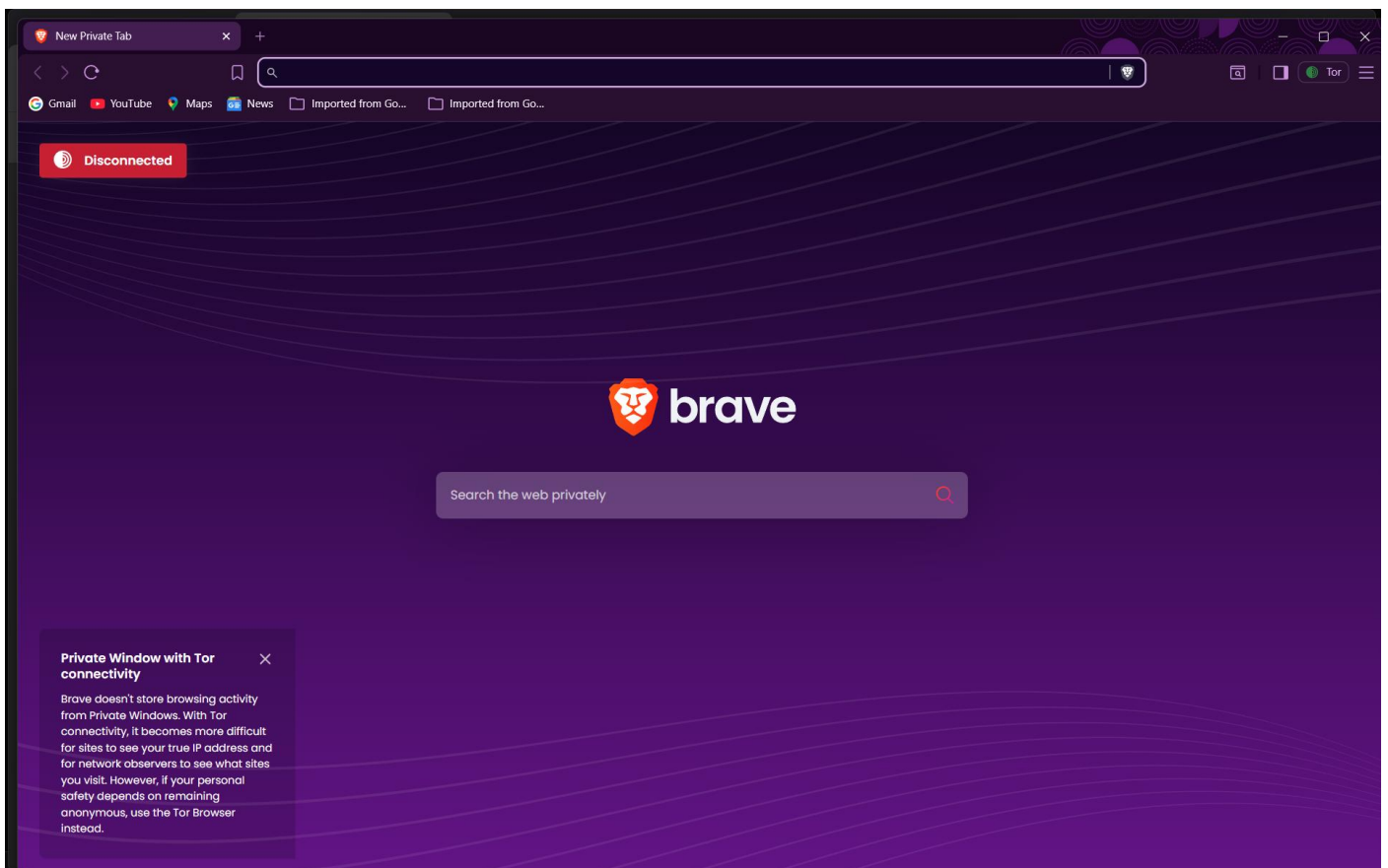
[Show more details](#)

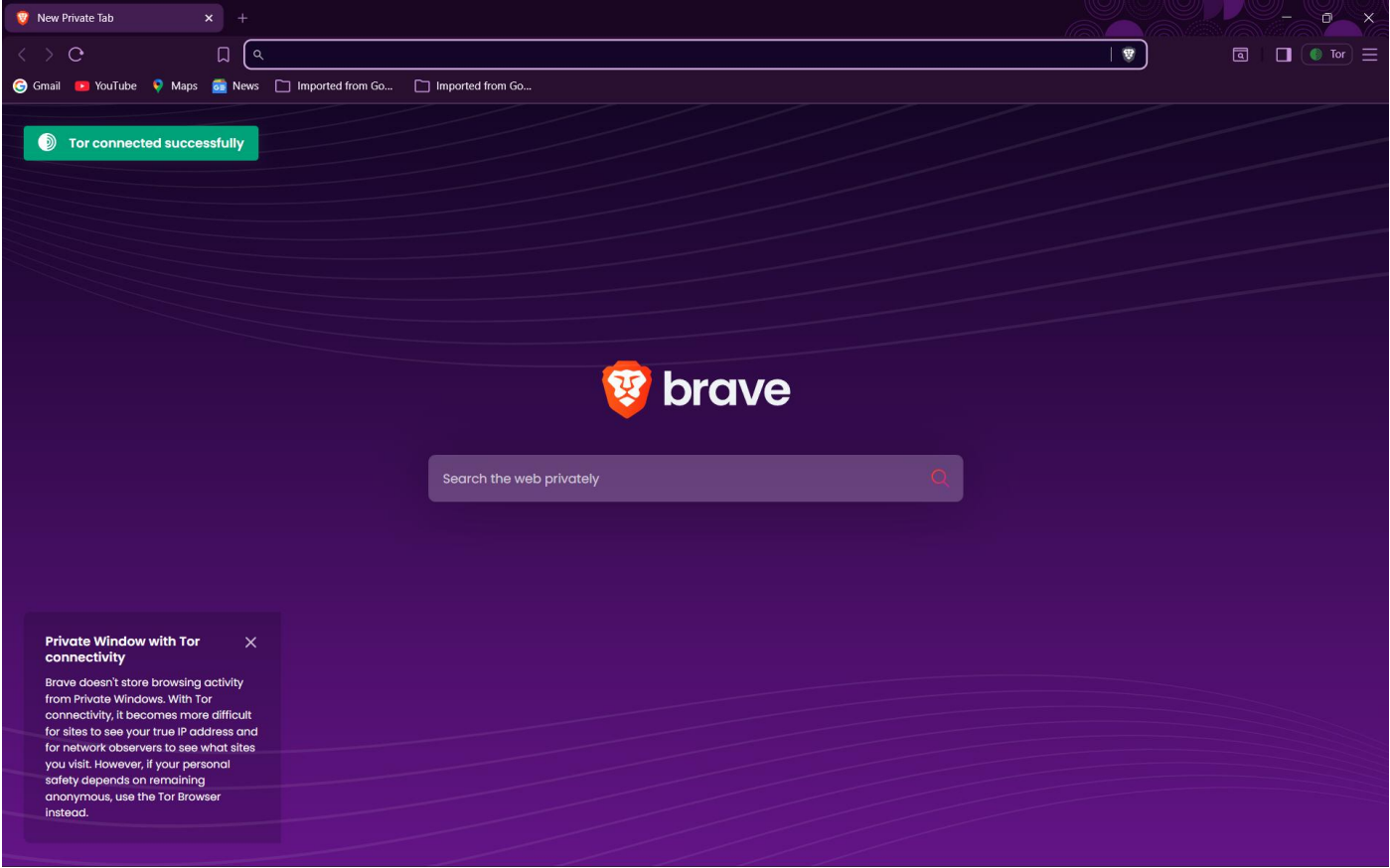
Yes

No



Tor: If Ip is blocked for the tor browser download but can be accessible to load it through brave





Assignment 7: Open the website, www.cert-in.org.in

- (i) write the procedure to report an incident.
- (ii) write the procedure to report about the vulnerabilities.
- (iii) Procedure to secure your PC.
- (iv) List down various security related tools and websites with proper explanation. Also install these tools on your system.

Procedure to Report an Incident to CERT-In

1. Choose the Reporting Method

An incident can be reported to CERT-In using any of the following channels:

- **Email:** incident@cert-in.org.in
 - **Phone (Helpdesk):** +91-1800-11-4949
 - **Fax:** +91-1800-11-6969
 - **Online Portal:**
<https://www.cert-in.org.in/SecurityIncident.jsp>
-

2. Reporting Timeline (6-Hour Rule)

- Incidents involving **data breaches, system compromise, or threats to human safety** must be reported **within 6 hours of detection**.
-

3. Information to Include in the Report

Provide as much accurate information as possible, including:

- Time and date of incident occurrence
 - Affected systems, servers, or networks
 - Observed symptoms or indicators of compromise
 - Security controls or safeguards in place
 - Immediate actions taken for containment or mitigation
-

4. CERT-In Review and Response

Upon receiving the report, CERT-In will:

- Verify the submitted information
- Assign a **tracking/reference number**
- Contact the reporting entity if additional details are required

CERT-In's response assistance includes:

- Incident identification and analysis
- Containment to limit further impact
- Eradication of the threat
- Recovery guidance to restore systems

Note: CERT-In provides advisory and coordination support but does **not** deploy personnel onsite.

5. Follow-Up Actions

- The organization may be requested to submit progress updates or technical logs
 - Additional forensic or system data may be required for deeper analysis
-

Procedure to Report a Vulnerability to CERT-In

Vulnerabilities can be responsibly disclosed through CERT-In's **RVDCP (Responsible Vulnerability Disclosure and Coordination Policy)** portal:

 **Vulnerability Reporting Portal:**
<https://www.cert-in.org.in/RVDCP.jsp>



The screenshot shows the CERT-In website with the header "Indian Computer Emergency Response Team" and "Ministry of Electronics and Information Technology, Government of India". The main navigation bar includes links like HOME, ABOUT CERT-In, KNOWLEDGEBASE, TRAINING, ADVISORIES, VULNERABILITY NOTES, and CYBER SECURITY ASSURANCE. The left sidebar contains various links such as Charter & Mission, Roles & Functions, Advisory Committee, Act/Rules/Regulations, Botnet Cleaning, Press, Tender, Contact Us, and links to various committees and partners like APCERT, TF-CERT, IAPWG, and CVE.

Home - Responsible Vulnerability Disclosure and Coordination Policy

Responsible Vulnerability Disclosure and Coordination Policy

1. Introduction

Responsible Vulnerability Disclosure and Coordination refers to the process of collection, analysis, mitigation coordination with researchers/finders and vendors leading to the public disclosure of newly identified cybersecurity vulnerabilities. The purpose of Responsible Vulnerability Disclosure and Coordination is to ensure that affected vendors/OEMs get sufficient time to remediate the vulnerability.

Indian Computer Emergency Response Team (CERT-In) collaborates with researchers, cybersecurity organizations, academic institutions, vendors/OEMs, and CERT's all over the world on handling of reported vulnerabilities. In this direction, CERT-In has formulated this Responsible Vulnerability Disclosure and Coordination Policy with an aim to strengthen trust in "Digital India", "Make in India" as well as to encourage responsible vulnerability research in the Country.

The policy outlines the process of vulnerability reporting to CERT-In and also provides details as to how the information provided to CERT-In will be handled and shared.

2. Reporting vulnerabilities to CERT-In

Security vulnerabilities in any product can be sent to us via email at vdisclose@cert-in.org.in. CERT-In accepts PGP Encrypted emails and attachments. The details of the public key are hereunder:

Email ID: vdisclose@cert-in.org.in
Key ID: 0x3B4E082C
Key Type: RSA
Expires Date: 2026-12-31
Key Size: 4096/4096
Fingerprint: 6927 2217 D8D4 0208 6B1C 23E9 CE29 EA67 3B4E 082C
Helpdesk: +91-1800-11-4949 (Toll Free)
Fax: +91-1800-11-6969

Acknowledgement will be sent within 72 working hours upon receipt of the vulnerability information by CERT-In.

Vulnerability Reporting Form

3. Details expected in vulnerability reports

In order to examine and validate the vulnerability, CERT-In will look forward to certain details as indicated below:

- The product(s) affected
- The exact software version or model affected;
- Vendor details
- Description of the vulnerability along with concise steps to reproduce the reported vulnerability along with supporting evidences such as:
 - Proof of concept (PoC) and/or
 - Code sample and/or
 - Crash reports and/or
 - Screenshots and Video recording etc.
- The impact of exploiting the vulnerability

In addition to the above, preferably the following details also may be provided:

- Other products or software versions likely to be affected
- How the vulnerability was discovered
- The tools used for discovering the vulnerability
- Information on any known exploit
- Time constraints with respect to going public about the issue (e.g. article, blog or conference etc.)
- Whether the vulnerability has already been reported to the vendor / other agency or any plan to do so
- Whether reporting party wants to remain anonymous during the coordination process
- Whether reporting party wants mention in the vulnerability note / advisory

Following need to be ensured before reporting the issue:

- The vulnerability must be reproducible on the latest available version or 'supported' version of the product
- The vulnerability must not be previously known

4. Coordination for resolution

CERT-In will examine and validate the vulnerability report and communicate to the discloser whether or not the report will be coordinated by CERT-In. Upon successful validation, CERT-In will initiate coordination with the relevant product vendor(s), discloser and other stakeholders (if required) for the remediation and closure of the issue.

CERT-In shall make all possible efforts to limit the disclosure to a bare minimum. However situations may arise where assistance from trusted third parties may be required in which case CERT-In will be sharing a subset or all the vulnerability information, as the case may be, with the trusted third parties.

CERT-In will also release vulnerability note/advisory on its website after the vulnerability is addressed or at an appropriate time as determined by CERT-In in synchronisation with the stakeholders involved during the coordination effort.

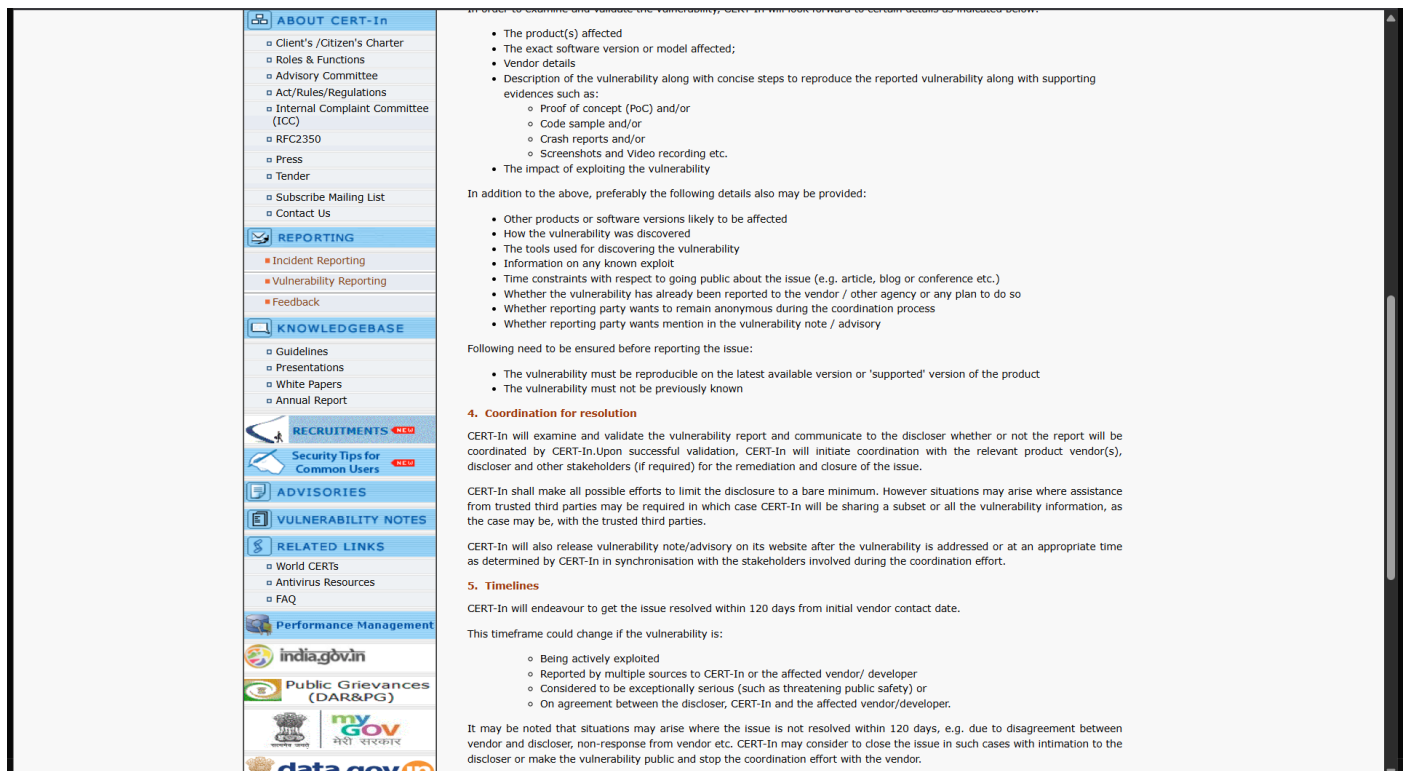
5. Timelines

CERT-In will endeavour to get the issue resolved within 120 days from initial vendor contact date.

This timeframe could change if the vulnerability is:

- Being actively exploited
- Reported by multiple sources to CERT-In or the affected vendor/ developer
- Considered to be exceptionally serious (such as threatening public safety) or
- On agreement between the discloser, CERT-In and the affected vendor/developer.

It may be noted that situations may arise where the issue is not resolved within 120 days, e.g. due to disagreement between vendor and discloser, non-response from vendor etc. CERT-In may consider to close the issue in such cases with intimation to the discloser or make the vulnerability public and stop the coordination effort with the vendor.



The screenshot shows the CERT-In website with the header "Indian Computer Emergency Response Team" and "Ministry of Electronics and Information Technology, Government of India". The main navigation bar includes links like HOME, ABOUT CERT-In, KNOWLEDGEBASE, TRAINING, ADVISORIES, VULNERABILITY NOTES, and CYBER SECURITY ASSURANCE. The left sidebar contains various links such as Charter & Mission, Roles & Functions, Advisory Committee, Act/Rules/Regulations, Botnet Cleaning, Press, Tender, Contact Us, and links to various committees and partners like APCERT, TF-CERT, IAPWG, and CVE.

ABOUT CERT-In

- Client's /Citizen's Charter
- Roles & Functions
- Advisory Committee
- Act/Rules/Regulations
- Internal Complaint Committee (ICC)
- RFC2350
- Press
- Tender
- Subscribe Mailing List
- Contact Us

REPORTING

- Incident Reporting
- Vulnerability Reporting
- Feedback

KNOWLEDGEBASE

- Guidelines
- Presentations
- White Papers
- Annual Report

RECRUITMENTS

Security Tips for Common Users

ADVISORIES

VULNERABILITY NOTES

RELATED LINKS

- World CERTs
- Antivirus Resources
- FAQ

Performance Management

india.gov.in

Public Grievances (DAR&PG)

myGov

data.gov

Common Users

ADVISORIES

VULNERABILITY NOTES

RELATED LINKS

- World CERTs
- Antivirus Resources
- FAQ

Performance Management

india.gov.in

Public Grievances (DAR&PG)

myGov

data.gov.in

ISAC Power

सामूहिक मददका सेवा पोर्टल

NATIONAL VULNERABILITY SERVICES PORTAL

Modes of Digital Payments

DIGITAL PAYMENT इुराक़्हा

RTI

Collaboration and Engaging with CERT-In

State/Sectoral CSIRT

disclosure and other stakeholders (if required) for the remediation and closure of the issue.

CERT-In shall make all possible efforts to limit the disclosure to a bare minimum. However situations may arise where assistance from trusted third parties may be required in which case CERT-In will be sharing a subset or all the vulnerability information, as the case may be, with the trusted third parties.

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6. Credits and Rewards

CERT-In appreciates the work of the vulnerability researcher(s) and is willing to give due credit for discovering the issue by mentioning the same on vulnerability note / advisory to be published on CERT-In website for the issue.

In addition, if the vendor(s) affected by the vulnerability desires to reward the discloser in some way, CERT-In will be willing to connect them with the discloser. CERT-In will limit its role to mere introducing the discloser with the vendor and will not partake in any discussion related to the reward whatsoever.

Vulnerability Hall of Fame

7. Disclaimer

CERT-In will make all attempts to expedite the vulnerability resolution, however no fixed deadlines can be assigned due to involvement of wide range of vendor(s), their issue handling processes, cooperation and priorities as well as uncertainties surrounding the complexity of solution which are well beyond the control of CERT-In. CERT-In will attempt to balance the interest of all stakeholders involved during the coordination effort.

This policy is subjected to IT Act 2000 and 2008 (Amendment). The reporting party must ensure to comply with all the extant laws and regulations while discovering the vulnerabilities. Reporting a vulnerability to CERT-In does not imply being exempt from compliance. Discloser shall be responsible for any action performed by her / him for discovering the vulnerability whatsoever.

"Install genuine and updated software to strengthen your online safety and security"

Indian Computer Emergency Response Team - CERT-In, Ministry of Electronics and Information Technology, Government of India.

Assignment 9: Install Hacker Process inside your PC and perform the demonstration.

Experiment 1: Select any browser and try to secure your browser by following settings:

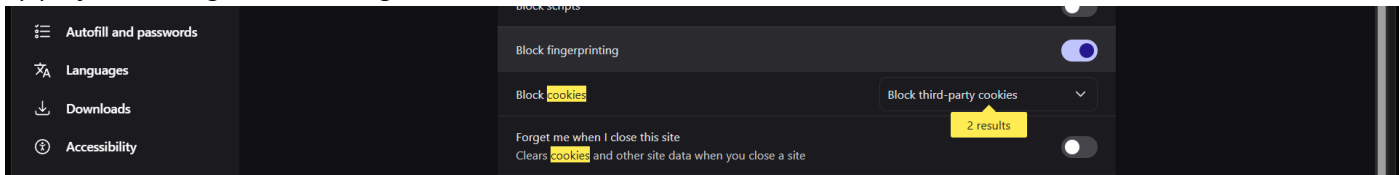
(i) trusted sited/blocked sites etc.

The screenshot shows the Brave browser settings page, specifically the 'Privacy and security' section. The left sidebar contains various settings categories, with 'Privacy and security' selected. The main content area lists several privacy and security settings, each with a toggle switch and a 'Learn more' link.

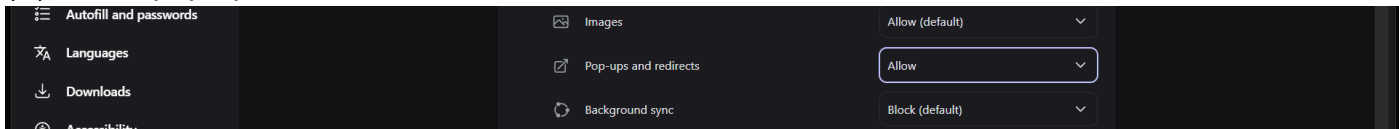
- Delete browsing data:** Delete history, cookies, cache, and more. (Toggle: On)
- Security:** Safe Browsing (protection from dangerous sites) and other security settings. (Toggle: On)
- Site and Shields Settings:** Controls what information sites can use and show (location, camera, pop-ups, and more). (Toggle: On)
- WebRTC IP handling policy:** Default. (Dropdown: Default)
- Use Google services for push messaging:** (Toggle: Off)
- Auto-redirect AMP pages:** Prefer publisher page URLs, instead of AMP and similar versions. (Toggle: On)
- Auto-redirect tracking URLs:** Enable support for bypassing top-level redirect tracking URLs. (Toggle: On)
- Prevent sites from fingerprinting me based on my language preferences:** Enabling this setting reduces how much web sites can learn about your language preferences. (Toggle: On)
- Block Microsoft Recall:** Block the Recall feature on Microsoft Windows computers from automatically taking screenshots of any open Brave tabs. (Toggle: On)
- Send a "Do Not Track" request with your browsing traffic:** Sites use their discretion when responding to this request. (Toggle: Off)

At the bottom of the page, there is a 'Tor windows' button.

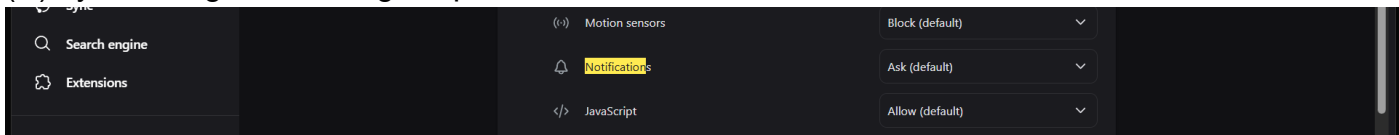
(ii) by enabling or disabling the cookies.



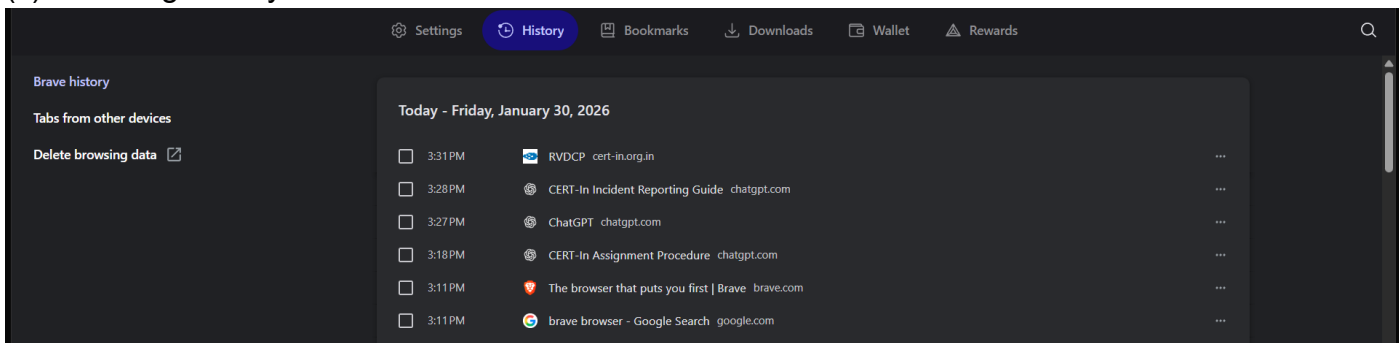
(iii) use of pop up blocker



(iv) by enabling or disabling scripts



(v) browsing history



(vi) saving passwords/master password

